

MACKINAC ISLAND, MI, USA

WMO: 727435

Lat: 45.865N

Lon: 84.637W

Elev: 740

StdP: 14.31

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 54820

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-4.2	1.1	-10.6	3.2	-2.7	-5.8	4.2	3.0	26.2	18.3	24.0	17.9	8.5	270	0.535

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	14.8	80.9	68.9	78.4	67.3	75.2	66.2	72.1	77.0	70.3	75.1	68.5	73.3	5.5	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
70.5	115.3	74.9	68.5	107.7	73.1	66.4	99.9	71.3	36.3	76.8	34.8	74.7	33.2	73.5	78.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
21.0	18.6	16.4	DB	-8.9	85.5	8.4	2.6	-14.9	87.4	-19.8	88.9	-24.5	90.4	-30.6	92.3	(4)
			WB	-9.5	74.6	8.2	1.8	-15.3	75.9	-20.1	76.9	-24.7	78.0	-30.6	79.3	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	43.8	20.0	20.6	28.3	39.1	50.4	60.5	66.5	65.7	60.2	48.3	37.5	27.6
	DBStd	18.27	10.53	10.17	9.66	7.54	7.03	6.21	5.13	4.95	6.95	7.84	7.82	8.45
	HDD50	4058	930	823	675	334	81	3	0	0	6	129	381	696
	HDD65	7944	1395	1243	1138	776	454	162	45	52	173	520	825	1161
	CDD50	1810	0	0	1	8	94	318	510	485	313	76	5	0
	CDD65	225	0	0	0	0	3	27	91	72	30	2	0	0
	CDH74	870	0	0	0	0	35	140	359	239	92	5	0	0
	CDH80	99	0	0	0	0	6	20	45	20	8	0	0	0
Wind	WSAvg	6.9	8.3	8.0	7.1	7.4	6.6	5.6	5.3	5.2	5.8	7.3	8.0	8.4
Precipitation	PrecAvg	30.6	1.6	1.1	1.7	2.4	3.0	3.0	2.9	3.0	3.5	3.8	2.7	1.9
	PrecMax	36.5	3.3	2.2	4.1	5.9	5.4	8.3	5.0	5.6	6.9	8.1	4.9	4.1
	PrecMin	22.8	0.3	0.2	0.1	1.0	0.8	0.8	1.3	0.7	1.5	0.9	0.5	0.2
	PrecStd	3.7	0.9	0.6	1.1	1.1	1.1	1.6	1.0	1.4	1.4	1.7	1.3	0.9
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	44.8	45.3	58.9	66.4	78.9	82.3	84.2	82.1	81.3	72.9	62.7	51.1
		MCWB	40.7	42.2	51.3	54.3	63.1	68.1	70.9	70.9	69.9	64.5	54.3	49.0
	2%	DB	38.7	39.1	48.5	60.0	71.7	78.5	81.0	79.2	76.6	67.7	55.1	43.6
		MCWB	36.6	36.7	43.6	48.8	60.2	66.3	68.7	68.5	66.9	60.5	50.5	40.7
	5%	DB	36.1	36.2	44.6	54.7	66.2	74.2	78.0	76.7	73.1	63.4	51.8	40.8
		MCWB	34.6	34.5	39.7	45.2	56.8	63.8	67.4	67.1	65.3	57.7	48.1	38.2
	10%	DB	33.8	34.0	39.9	50.3	63.0	71.7	75.2	74.0	71.0	59.7	47.9	38.0
		MCWB	32.7	32.5	36.2	42.6	54.5	62.4	66.6	66.1	64.4	54.9	44.9	36.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	42.0	42.3	52.6	57.5	66.7	73.0	74.7	74.4	72.2	67.9	55.4	48.9
		MCDB	43.8	44.4	56.1	63.1	72.8	79.1	79.9	79.7	77.0	71.1	60.9	50.1
	2%	WB	37.0	37.2	45.3	50.3	62.3	68.8	72.3	71.5	69.5	62.2	51.7	42.1
		MCDB	38.6	38.5	48.0	57.5	68.7	74.7	77.1	75.8	74.0	64.8	54.4	43.7
	5%	WB	34.6	34.5	40.0	46.4	59.0	66.3	70.4	69.7	67.4	58.3	48.0	38.7
		MCDB	35.8	35.6	43.3	52.1	64.7	71.3	75.2	74.2	71.6	62.7	50.5	40.4
	10%	WB	32.4	32.3	36.9	43.5	55.4	63.8	68.6	67.9	65.5	55.5	45.5	36.6
		MCDB	33.4	33.5	40.0	49.4	61.4	69.0	73.4	72.7	69.4	59.2	47.4	38.1
Mean Daily Temperature Range	5% DB	MDBR	11.0	12.3	13.3	14.2	16.4	15.8	14.8	14.0	13.7	11.4	9.5	9.2
		MCDBR	10.9	12.4	16.6	22.3	22.7	19.7	16.7	15.9	16.0	15.3	14.1	11.4
		MCWBR	9.8	10.7	12.2	13.8	13.5	11.1	9.1	9.1	9.6	11.0	11.6	10.2
	5% WB	MCDBR	10.5	11.6	15.8	20.3	20.0	16.0	14.0	13.3	14.3	14.3	13.1	10.4
	MCWBR	9.8	10.5	12.3	14.1	14.0	11.0	9.2	9.3	10.2	11.4	11.4	9.9	
Clear-Sky Solar Irradiance	taub	0.298	0.292	0.316	0.357	0.372	0.390	0.389	0.380	0.367	0.333	0.301	0.279	
	taud	2.331	2.361	2.398	2.335	2.371	2.355	2.379	2.402	2.427	2.503	2.522	2.457	
	Ebn at Noon	253	282	291	286	284	278	277	275	270	265	255	252	
	Edh at Noon	24	29	32	37	37	38	37	35	31	25	20	19	
All-Sky Solar Radiation	RadAvg	356	650	1089	1448	1786	1933	1945	1649	1247	725	382	267	
	RadStd	39	67	116	154	149	137	85	104	100	84	53	25	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (38 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page